SANDEEP UTHAYAKUMAR

Software Engineer | HSBC

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PROFILE

Software Engineer at HSBC working on trading floor systems and the Verint compliance platform, with a background in backend API development, full-stack web engineering and applied deep learning. Published researcher (IEEE, ScienceDirect) with a record of delivering measurable performance gains.

EXPERIENCE

HSBC — Software Engineer **Jul 2026 – Present**

Pune, India

- Develop and support trading floor applications for front-office desks, maintaining availability of latency-sensitive systems throughout market hours.
- Work across the Verint platform on workforce management and analytics, including integration and migration work within the trading environment.

VidyaInternaHub — Backend Developer Intern **May 2025 – Jul 2025**

Remote, India

- Re-engineered CRM backend APIs to improve server-side stability, reducing average response time by 70%.
- Optimised request handling so the service sustained approximately 75,000 requests per hour without degradation.

National Institute of Technology Puducherry — Research Intern **Sep 2023 – Dec 2023**

Karaikal, India | Team of 3

- Built a Python cryptocurrency wallet with blockchain integration — account creation, balance viewing and transactions through a PyQt5 GUI — implementing the core ledger and proof-of-work transaction validation. [GitHub]

EDUCATION

B.Tech — National Institute of Technology Puducherry **2022 – 2026 · CGPA 8.00/10**

Senior Secondary (CBSE) — Aditya Vidyashram, Gurugram **2022 · 85.80%**

Secondary (CBSE) — Amrita Vidyalayam **2020 · 91.40%**

PROJECTS

Sicko Shoe Store — Next.js E-Commerce Platform **Oct 2025**

Next.js · React · Tailwind CSS

- Built a responsive, SEO-optimised storefront using Next.js server-side rendering for faster page loads and stronger search visibility.
- Implemented dynamic product listings with interactive filtering and sorting.

APK Malware Detection & Family Prediction — CNN-LSTM + Random Forest **Aug 2024 – Nov 2024**

Python · TensorFlow · scikit-learn

- Designed a hybrid CNN-LSTM pipeline with Random Forest classifiers to detect Android malware and predict its family, reaching a 99.98% F1-score against published baselines.

Student-GPT — Transformer Q&A for DSA **Jul 2024 – Aug 2024**

React · FastAPI · Python · PyTorch

- Trained a Transformer model to answer Data Structures & Algorithms queries in natural language, served through a React and FastAPI web interface.

PUBLICATIONS

APK Malware Detection and Family Prediction Using CNN-LSTM and RF Classifiers **Jul 2025**

IEEE · 16th ICCCNT 2025

- Reshaped 215 static features into a 43×5 matrix; CNN layers extract spatial features while LSTM layers capture sequential patterns, followed by dropout and dense layers for binary classification.
- Reported a 99.98% F1-score, exceeding existing benchmark models.

Multi-Head Attention Transformer for Text-to-Text Translation **May 2025**

ScienceDirect

- Proposed a multi-head attention Transformer for English-to-Tamil translation, achieving a 40% improvement in translation accuracy over the baseline.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Next.js, React, Node.js, Express, Angular, FastAPI, PyTorch, TensorFlow

Databases & Tools: MySQL, MongoDB, Git, GitHub, IntelliJ IDEA, VS Code, Postman